

AI Powered Stock Market Insight Generator for the Colombo Stock Exchange

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Project Proposal Report

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Component: Insight Visualization and Human Interpretability Evaluation

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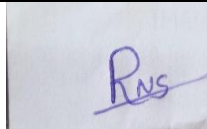
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The Colombo Stock Exchange (CSE) has experienced increasing participation from retail investors, particularly younger individuals with limited financial literacy. However, AI-generated financial insights such as machine learning trend outputs, explainable AI interpretations, large language model summaries, and sentiment-driven market signals remain difficult for non-expert investors to understand without suitable visualization and explanation support. Existing investor platforms primarily focus on technical charting and raw financial data presentation, while academic work on explainable AI in finance rarely validates whether such explanations are understandable to end users. This creates a gap in investor-facing, interpretable financial insight systems for the CSE context.

This research addresses that gap through an **Insight Visualization and Human Interpretability Evaluation** component within a broader AI-powered stock market insight generator. The component integrates outputs from upstream AI subsystems into an interactive dashboard and compares multiple presentation strategies, including chart-only views, chart-plus-text explanations, SHAP-based feature contribution visualizations, sentiment-event indicators, financial summary cards, and composite risk warnings. Human interpretability is evaluated using the System Usability Scale (SUS), a custom Financial Insight Comprehension Score (FICS), adapted NASA-TLX, task completion measures, and think-aloud analysis. The system is implemented using React.js, Node.js, MongoDB Atlas, Recharts, and D3.js.

The expected outcome is an empirically validated human-centered visualization framework and a refined dashboard prototype that improves comprehension, trust, usability, and manageable cognitive load for non-expert CSE retail investors. Socially, the research supports financial literacy and responsible investor participation. Commercially, it can be adapted for brokerage platforms, fintech applications, and investor education services.

Keywords: Insight Visualization, Human Interpretability Evaluation, Explainable AI, Colombo Stock Exchange, Investor Dashboard

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
LLM	Large Language Model
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
API	Application Programming Interface
UI	User Interface
UX	User Experience
SUS	System Usability Scale
FICS	Financial Insight Comprehension Score
CSE	Colombo Stock Exchange
IT	Information Technology
SLIIT	Sri Lanka Institute of Information Technology
WBS	Work Breakdown Structure
MERN	MongoDB, Express, React, Node.js
REST	Representational State Transfer
NASA-TLX	NASA Task Load Index
SDG	Sustainable Development Goal

INTRODUCTION

1.1 Background and Context

Stock markets play an important role in the financial system by helping companies raise capital while giving individuals and institutions opportunities to invest and build wealth. In Sri Lanka, the Colombo Stock Exchange (CSE) is the main platform for trading publicly listed company shares. Over the last decade, retail participation in the CSE has increased due to online trading platforms, wider internet access, and growing interest in alternative investment opportunities. However, many retail investors still find it difficult to analyze market information in a meaningful way because understanding stock market behaviour requires more than simply looking at prices; investors often need to interpret trading volumes, volatility, technical indicators, company disclosures, and market sentiment. For users without a strong financial background, this information can be overwhelming.

This problem mainly affects **non-expert retail investors**, especially younger and digitally active users who increasingly depend on mobile trading platforms, social media discussions, and informal market narratives. The TAF states that the rise in retail participation has been accompanied by misinformation, emotionally driven decision-making, and unverified advisory content, which can mislead inexperienced investors and increase vulnerability to poor decisions, reduced market confidence, and even fraudulent schemes. It also notes that retail investors struggle to interpret market data, annual reports, macroeconomic indicators, and geopolitical news without proper tools that explain risk, trends, and contextual signals.

The importance of this problem is both **economic and social**. Reduced investor confidence can negatively affect liquidity and capital formation, while misinformation can fuel speculative behaviour and weaken the healthy functioning of the capital market ecosystem. The TAF explicitly frames this issue as having broader economic and social implications and argues for an AI-driven system that helps investors understand **why** market changes occur without functioning as a direct financial advisory service. Such a system is expected to promote responsible participation and improve confidence in the market.

Existing solutions partially address this need, but they remain insufficient. Commercial platforms such as TradingView, MetaTrader, Yahoo Finance, and Investing.com provide technical indicators, charting tools, and screening features, but they assume prior financial knowledge, do not provide explainable AI-driven

insights, and are not optimized for emerging market characteristics such as those of the CSE. They also lack contextual interpretations based on macroeconomic events, geopolitical news, and financial document understanding relevant to the Sri Lankan market. Academic research similarly focuses heavily on predictive performance rather than interpretability or investor education, meaning that users may receive directional predictions without understanding the underlying drivers or influencing factors. While SHAP and LIME have advanced model transparency, they have not been widely applied to investor-facing financial dashboards in emerging market settings.

To address these limitations, the broader project proposes an **AI Powered Stock Market Insight Generator** for the Colombo Stock Exchange. The system ingests three categories of data: historical stock price and trading volume data, financial documents such as annual reports and earnings statements, and news articles with macroeconomic information. These structured, semi-structured, and unstructured data sources are processed through separate analytical components before being exposed through an investor-facing interface. Within that larger solution, the **Insight Visualization and Human Interpretability Evaluation** component acts as the final layer of the pipeline, converting outputs from the three upstream AI subsystems into an interactive dashboard that is empirically validated for interpretability. The TAF defines this component as responsible for visualizing trends, sentiment signals, and explanations; integrating ML, sentiment, and LLM outputs into a unified interface; conducting interpretability evaluation; and refining the interface through feedback-driven usability improvement.

Accordingly, this research focuses on how AI-generated financial insights should be visually presented and explained so that non-expert investors can understand them. Rather than treating the dashboard as a simple interface implementation, the component investigates a human-centered framework that combines insight visualization, explanation support, and structured interpretability evaluation using instruments such as SUS, FICS, and NASA-TLX. The current proposal already positions this component around a React.js dashboard with five specialized panels, a fusion layer that standardizes upstream outputs, and a formal evaluation module supported by formative and summative testing.

This research is also aligned with **Sustainable Development Goal 8 (Decent Work and Economic Growth)** by supporting informed financial participation and healthier capital market behaviour, and with **Sustainable Development Goal 9 (Industry, Innovation and Infrastructure)** by promoting responsible AI adoption in a critical financial domain. Both the TAF and the current proposal explicitly identify SDG 8 and SDG 9 as relevant to the project.

From a research-cluster perspective, this work belongs to the **AI-powered financial analytics and human-centered decision-support systems** domain. It sits at the intersection of information visualization, explainable AI, human-computer interaction, and financial technology, while contributing a localized CSE-specific investor support framework rather than a generic global-market dashboard. This is consistent with the TAF's research-group framing under the AI-powered stock market insight generator and the specific novelty assigned to this member's component: **human-centered interpretability evaluation of AI-generated financial insights enabling effective investor comprehension through visualization and usability-driven refinement.**

1.2 Literature Review

Financial visualization systems and investor platforms have long been used to support decision-making through charts, indicators, dashboards, and summarized market data. Commercial platforms such as TradingView, Yahoo Finance, and MetaTrader provide strong charting and monitoring capabilities, allowing users to observe price movements, trading volume, and technical indicators. However, these systems are primarily designed for users who already possess financial knowledge and chart-reading experience. They focus more on data presentation than on explanation, and they rarely provide AI-generated insight in a form that is easily understandable to non-expert retail investors. Research on investor behaviour also shows that individual investors often rely on incomplete reasoning, narratives, and sentiment rather than systematic evidence-based interpretation, which increases the importance of clear and user-friendly market explanation [1], [2], [3]

In parallel, Explainable Artificial Intelligence (XAI) has emerged as an important area in machine learning research. Studies by Molnar [4], Doshi-Velez and Kim [5], and Lundberg and Lee [7] emphasize the need to make model outputs understandable, transparent, and trustworthy. Among available explanation methods, SHAP has become especially influential because it quantifies feature contribution in a mathematically grounded manner. These approaches are highly useful from a technical perspective because they help explain why a model produced a particular output. However, most XAI studies evaluate explanation quality at the algorithmic level rather than at the end-user level. As a result, there is still limited evidence on whether non-expert investors can actually understand

SHAP-based or feature-level explanations when they are displayed through financial dashboards.

Research in information visualization provides additional guidance on how quantitative information should be presented to users. Few [8], Ware [9], and Munzner [10] argue that effective visualization depends not only on accuracy, but also on perception, context, visual encoding, and cognitive effort. These studies suggest that users interpret information more effectively when visual structures are aligned with their goals and mental models. In financial applications, this means that complex charts alone may not be sufficient for novice or intermediate investors. Supporting elements such as annotations, simplified summaries, contextual explanations, risk indicators, and event-linked visual cues may improve interpretability and reduce confusion.

Human-computer interaction research further highlights the importance of evaluating usability and mental workload when designing information systems. Standard frameworks such as the System Usability Scale (SUS) [5] and NASA-TLX [6] are widely used to evaluate interface usability and perceived cognitive load, while broader interface design principles emphasize consistency, simplicity, and task-oriented interaction [17]. These measures are useful for understanding whether a system is easy to use, but they do not directly determine whether users truly understand the financial meaning of the insights shown to them. This limitation is particularly important in investor-facing AI systems, where usability alone does not guarantee interpretability.

Overall, the reviewed literature and existing financial products show that significant progress has been made in stock market analytics, explainable AI, and information visualization. Commercial platforms provide strong charting and monitoring capabilities, while academic studies contribute methods for prediction, feature-level explanation, and usability assessment. However, these approaches remain limited in several ways. Existing investor platforms mainly emphasize technical charting and raw data presentation rather than user-friendly explanation, while most explainable AI research focuses on model transparency rather than whether end users can meaningfully understand the displayed explanations. In addition, currently available systems rarely integrate trend analysis, sentiment context, financial summary interpretation, and explanation support into one unified investor-facing interface, especially for a localized setting such as the Colombo Stock Exchange. These limitations indicate that there is still a need for a human-centered, explanation-oriented financial insight visualization framework that not only presents AI-generated outputs, but also validates whether those outputs are actually interpretable for non-expert retail investors. This unresolved need directly leads to the research gap addressed in the next section.

1.3 Research Gap

Although AI-based financial analysis has advanced considerably, an important gap remains between **generating financial insights** and **making those insights understandable to non-expert retail investors**. Existing commercial platforms such as TradingView, Yahoo Finance, and MetaTrader mainly provide technical charts, indicators, and raw financial data. While these platforms are useful for experienced traders, they are not designed to explain AI-generated outputs in a way that is easy for beginner or intermediate investors to understand. This means that current systems support data access, but not necessarily **insight comprehension**. For retail investors in the Colombo Stock Exchange (CSE), this problem becomes more significant because local investors often require simpler, contextual, and more interpretable presentation of market information.

A second gap lies in the current academic treatment of explainability. Existing research in explainable AI mainly focuses on improving **model-level transparency** through methods such as SHAP, feature importance, or explanation techniques. However, these approaches usually evaluate whether the model can be explained mathematically, rather than whether the intended end user can understand the explanation in practice. In other words, even if an AI model is technically explainable, the explanation may still be too complex, abstract, or visually difficult for a retail investor to interpret correctly. Therefore, current approaches are insufficient because they emphasize **technical interpretability** rather than **human interpretability**.

Another important gap is that most investor dashboards are designed as **fixed interfaces**. They typically present one standard visualization layout and assume that the displayed format is adequate for all users and all types of insight. However, the same financial insight may be understood differently depending on how it is presented. For example, a stock movement can be shown as a chart only, a chart with plain-language explanation, a chart with feature contribution, or a chart with sentiment and event context. Existing approaches do not systematically compare these alternative visualization and explanation strategies to determine which presentation pattern best improves comprehension, trust, and usability. This is a major limitation because visualization is not only a display task; it is also a communication task.

A further gap is the lack of a **unified investor-facing framework** tailored to the Colombo Stock Exchange context. Most existing systems are designed for global markets and do not reflect the local behavior, literacy levels, and interpretability needs of Sri Lankan retail investors. In addition, available systems do not combine multiple AI outputs such as trend analysis, XAI explanations, financial report summaries, and sentiment/event signals into one coherent and user-friendly dashboard. This shows that there is still no

localized framework that transforms multi-modal AI outputs into a single, understandable investor-facing insight layer.

There is also a clear gap in **evaluation methodology**. Standard usability measures such as SUS and NASA-TLX are helpful for measuring usability and cognitive load, but they do not directly measure whether the user actually understands the financial meaning of the displayed insight. A dashboard may be easy to use but still fail to communicate the intended financial explanation. Therefore, domain-specific interpretability measures such as the Financial Insight Comprehension Score (FICS) are needed to assess whether non-expert investors correctly understand stock trends, risk warnings, explanation panels, and summarized financial signals.

Therefore, the missing contribution is not simply another stock dashboard. The actual research gap is the absence of a **human-centered, evidence-based, and CSE-specific insight visualization framework** that:

- integrates multi-modal AI-generated outputs into one investor-facing dashboard,
- compares different visualization and explanation strategies for the same financial insight,
- evaluates investor understanding through structured interpretability measures,
- and refines the interface based on empirical user feedback.

Hence, the novelty of this research lies in transforming **Insight Visualization and Human Interpretability Evaluation** from a simple dashboard implementation task into a comparative, human-centered research framework that identifies how AI-generated financial insights should be visualized and explained so that non-expert CSE retail investors can genuinely understand them.

Table 1 - Research Gap Table

Function	Research Paper A	Research Paper B	Research Paper C	Proposed System
Financial trend visualization	✓	✓	✓	✓
AI-generated stock insight presentation	X	✓	X	✓
Integration of multiple insight sources (trend, sentiment, reports, XAI)	X	✓	X	✓

Function	Research Paper A	Research Paper B	Research Paper C	Proposed System
Explainable AI visualization for investor use	X	X	✓	✓
Multiple explanation strategies for the same insight	X	X	X	✓
Human interpretability evaluation	X	X	X	✓
Investor comprehension measurement (e.g., FICS)	X	X	X	✓
Usability and cognitive load evaluation	X	✓	X	✓
Iterative refinement based on user feedback	X	X	X	✓
Support for non-expert retail investors	X	X	X	✓
CSE-specific localized investor dashboard	X	X	X	✓

OBJECTIVE

2.1 Main Objective

To develop and validate a human-centered insight visualization and explanation framework that integrates multi-modal AI-generated financial outputs into an investor-facing dashboard, and to comparatively evaluate alternative presentation strategies in order to identify the most effective formats for improving comprehension, trust, usability, and cognitive accessibility among non-expert Colombo Stock Exchange retail investors.

2.2 Specific Objectives

1. To integrate outputs from market trend analysis, explainable AI, financial document understanding, and sentiment/event analysis into a unified dashboard interface.
2. To design and implement interactive visualization modules for stock trends, sentiment signals, risk indicators, financial summaries, and explanation panels.
3. To develop multiple visualization and explanation variants for the same financial insight, such as chart-only, chart-plus-text, chart-plus-feature-contribution, and chart-plus-event-context formats.
4. To evaluate investor comprehension, usability, trust, and cognitive load using SUS, FICS, adapted NASA-TLX, task completion measures, and think-aloud analysis.
5. To compare alternative presentation strategies and identify the most interpretable insight presentation patterns for non-expert investors.
6. To iteratively refine the dashboard and propose a human-centered insight visualization framework for the Colombo Stock Exchange context.

METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research approach because the goal is not only to analyze a problem, but also to design, implement, and evaluate a practical artefact. The artefact developed in this research is an investor-facing insight visualization and interpretability evaluation framework that presents AI-generated financial insights in a clear and explainable form.

3.2 Technical methods and algorithms

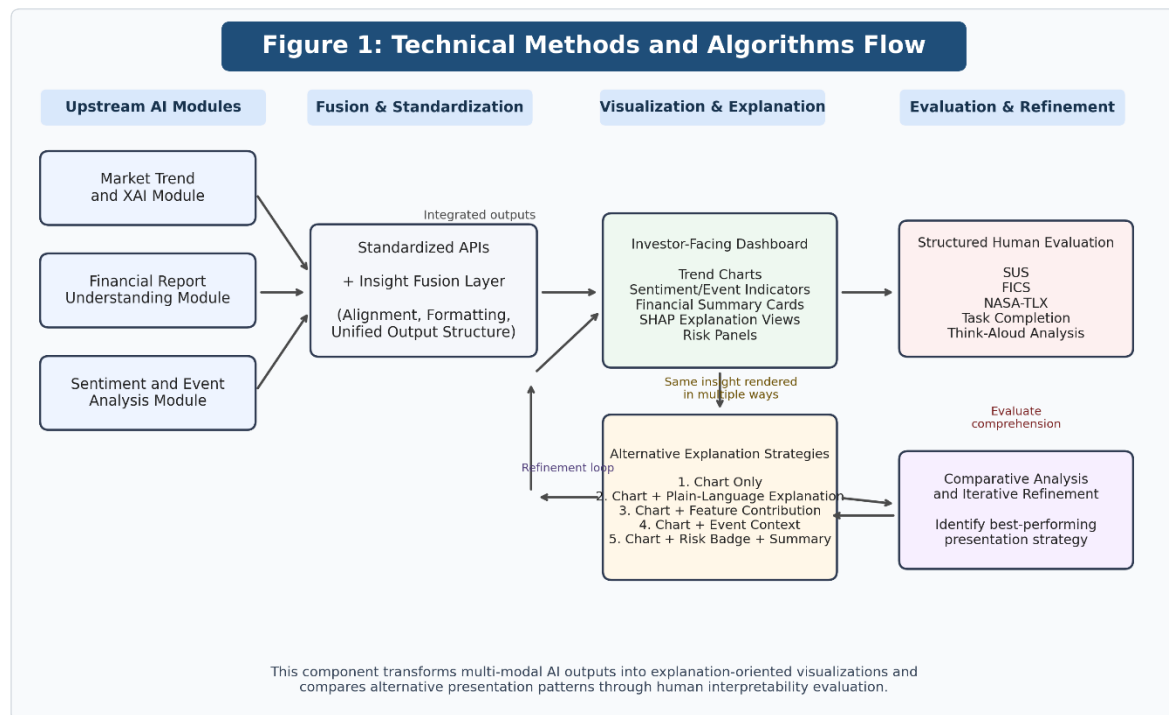


Figure 3

The proposed component operates as the final presentation layer of a broader AI-powered stock market insight generator. Outputs from upstream modules are received through standardized APIs and transformed into investor-facing visualizations. The dashboard includes stock trend charts, sentiment-event indicators, financial summary cards, SHAP-based explanation views, and composite risk panels.

To strengthen the research contribution, the study will not treat visualization as a single fixed dashboard. Instead, the same financial insight will be presented through multiple alternative explanation strategies, including:

- chart only
- chart + plain-language explanation
- chart + feature contribution
- chart + event context
- chart + risk badge + contextual summary

These variants will then be compared through structured human evaluation to identify which format most effectively supports investor understanding.

3.3 Tools and Platforms

A combination of modern web development, visualization, prototyping, and evaluation tools will be used to implement and validate the proposed Insight Visualization and Human Interpretability Evaluation component. Since this component functions as the investor-facing layer of the broader system, the selected tools must support both technical integration and user-centered interface design.

React.js will be used as the primary frontend framework for dashboard development. It supports modular component-based architecture, which is suitable for implementing multiple visualization panels such as stock trend charts, sentiment indicators, financial summary cards, SHAP-based explanation views, and risk warning banners. Its flexibility also allows the development of multiple visualization and explanation variants for comparative testing.

Node.js with Express.js will be used to build integration APIs and backend service endpoints. This layer will act as a middleware between upstream AI modules and the investor-facing dashboard by receiving, standardizing, and delivering insight data in a consistent structure. It is particularly useful for handling REST-based communication, response formatting, and integration of multi-modal outputs into a unified insight delivery pipeline.

MongoDB Atlas will be used for structured cloud-based data storage. It will store integrated outputs received from upstream modules, dashboard-ready processed records, evaluation responses, and experimental datasets. As a NoSQL database,

MongoDB is suitable for handling heterogeneous and semi-structured data produced by trend analysis, explanation outputs, sentiment signals, and financial summaries.

Recharts and D3.js will be used for data visualization. Recharts will be applied to build standard investor-facing dashboard elements such as line charts, bar charts, pie charts, and trend panels in a developer-friendly way. D3.js will be used where more advanced and customized visualizations are required, such as sentiment-event timelines, feature-contribution views, or interactive explanation overlays. Using both libraries enables a balance between rapid implementation and visualization flexibility.

Python will be used for preprocessing, formatting, and transformation support where necessary. Although the main dashboard implementation is web-based, Python can assist with intermediate tasks such as JSON restructuring, preprocessing of evaluation results, metric calculation, and generation of intermediate data representations required for visualization. It may also support analytical scripts used for computing measures such as FICS scores or summary statistics from user evaluation data.

Figma will be used for wireframing, interface mockup creation, and early-stage design iteration. Before full implementation, low-fidelity and high-fidelity mockups will be prepared in Figma to explore multiple dashboard layouts and explanation strategies. This supports user-centered design by enabling rapid comparison of interface concepts before development begins.

Google Forms will be used to collect structured evaluation responses from participants during formative and summative user studies. It will support the administration of the System Usability Scale (SUS), Financial Insight Comprehension Score (FICS) task responses, adapted NASA-TLX items, and additional feedback questions related to trust, clarity, and usefulness.

GitHub will be used for version control, collaborative development, and project management. It will help track code changes, maintain different development branches for visualization variants, and ensure proper documentation of implementation progress. GitHub also supports reproducibility and organized collaboration throughout the research lifecycle.

Overall, the selected tools and platforms were chosen because they collectively support the three major needs of this component:

1. interactive dashboard implementation,
2. multi-modal data integration, and
3. human-centered evaluation and iterative refinement.

These technologies are cost-effective, widely adopted, scalable, and suitable for developing a research-oriented financial insight visualization framework within the scope of an academic final-year project.

3.4 Data collection and ethical considerations

The component primarily uses outputs from the broader project’s AI modules, including market modelling, financial report understanding, and sentiment analysis. Additional data will be collected from usability and interpretability evaluation participants. Since the system is designed as a decision-support and educational tool rather than a financial advisory platform, participants will be clearly informed that the dashboard does not provide guaranteed investment recommendations. User responses will be anonymized, securely stored, and used only for academic evaluation purposes.

3.5 Validation Strategy and Performance Measurements

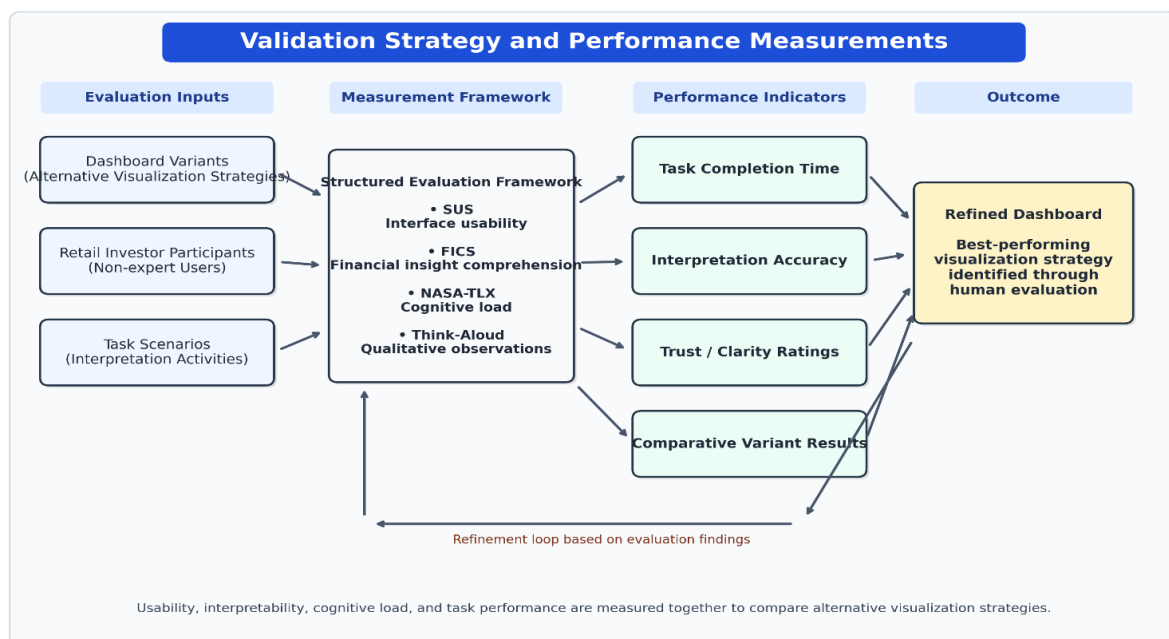


Figure 2

The component will be evaluated using both usability and interpretability measures. System Usability Scale (SUS) will assess interface usability. A custom Financial Insight Comprehension Score (FICS) will measure whether users correctly understand the presented financial insights. Adapted NASA-TLX will measure perceived cognitive load. Additional metrics such as task completion time, interpretation accuracy, trust ratings, and think-aloud observations will be used to compare alternative visualization strategies. The final dashboard will be refined based on these findings.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

The proposed solution functions as part of a broader **AI Powered Stock Market Insight Generator** for the **Colombo Stock Exchange (CSE)**. The system follows a layered workflow that begins with upstream AI analysis modules and ends with an investor-facing dashboard.

The architecture consists of five main layers. First, the **Upstream AI Analysis Modules** generate outputs related to market trends and XAI explanations, financial report understanding, and sentiment-event analysis. Second, the **Insight Fusion Layer** collects and standardizes these outputs into a common investor-facing format. Third, the **Insight Visualization Layer** presents the fused insights through trend charts, risk indicators, sentiment-event timelines, financial summary cards, and SHAP-based explanation views. Fourth, the **Explanation Strategy Layer** supports multiple presentation styles for the same insight, such as chart-only, chart-plus-text, and chart-plus-context views. Finally, the **Human Interpretability Evaluation Layer** collects SUS, FICS, NASA-TLX, task accuracy, and think-aloud data to evaluate user understanding and support refinement of the dashboard.

This modular architecture ensures that the component is not treated only as a dashboard interface, but as an empirically validated explanation framework for non-expert investors.

4.2 System Diagram of Components

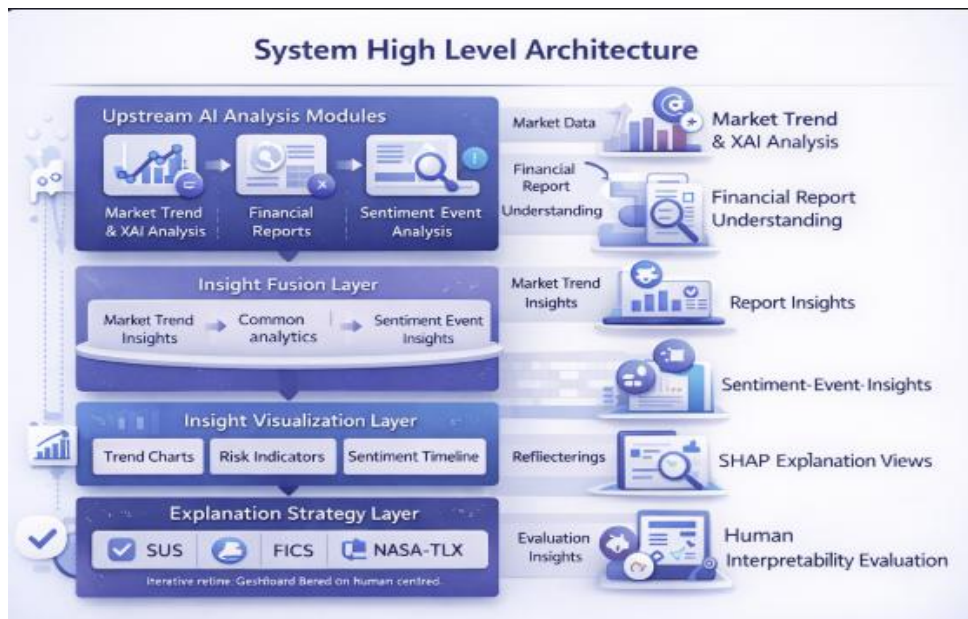


Figure 1



Figure 2

The system architecture is illustrated in **Figure 1** and **Figure 2**.

Figure 1 presents the layered structure of the proposed system, starting from the AI analysis modules and ending with human interpretability evaluation.

Figure 2 shows the overall system overview, where integrated insights flow into the investor dashboard and are later evaluated through structured human-centered methods.

Together, these figures show how multiple AI-generated outputs are transformed into investor-friendly visual explanations and how evaluation findings are used to improve the system.

4.3 Individual Component and Its Internal Parts

The **Insight Visualization and Human Interpretability Evaluation** component includes the following internal parts:

- **Upstream AI Input Reception** – receives outputs from market trend analysis, financial report understanding, and sentiment-event analysis modules.
- **Insight Fusion Layer** – standardizes and aligns outputs into a unified format.
- **Insight Visualization Layer** – generates charts, summaries, indicators, and explanation views.
- **Explanation Strategy Layer** – supports multiple visualization and explanation variants for the same insight.
- **Human Interpretability Evaluation Layer** – measures usability, comprehension, and cognitive load through SUS, FICS, NASA-TLX, and think-aloud analysis.
- **Iterative Refinement Layer** – improves the dashboard based on evaluation findings.

4.4 Component Interactions

The component interacts with all upstream analytical modules. Trend and XAI outputs, financial report summaries, and sentiment-event results are first collected and fused. These are then visualized in the investor dashboard using different explanation strategies. During evaluation, user responses are recorded and analyzed to identify which presentation formats are most understandable. The results are then used to refine the dashboard.

4.5 Data Flow Explanation

The data flow begins with three main input sources:

1. historical stock market data,
2. financial reports,
3. news and macroeconomic information.

These are processed by the corresponding AI modules and sent to the **Insight Fusion Layer**. The fused outputs are then passed into the **Insight Visualization Layer**, where they are displayed through charts, summaries, indicators, and explanation views. Multiple explanation strategies are then applied through the **Explanation Strategy Layer**. Finally, users interact with the dashboard, and their interpretability results are collected by the **Human Interpretability Evaluation Layer** to support refinement.

4.6 Scalability and Security Considerations

The proposed architecture is modular and scalable, allowing new insight types, data sources, and visualization panels to be added in the future. Since each layer is separated logically, the system can be maintained and expanded without major redesign.

From a security perspective, user responses and stored outputs should be protected through secure storage and controlled access. In addition, evaluation data should be anonymized and used only for academic purposes. The system is designed as an investor support tool, not as a direct financial advisory platform.

Overall, the high-level system architecture shows how multi-modal AI-generated financial outputs are transformed into a unified, investor-friendly, and empirically evaluated dashboard. By combining visualization, explanation strategies, and interpretability assessment, the proposed component provides both a technical solution and a human-centered research contribution.

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USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves multiple stakeholder groups who directly or indirectly interact with the Insight Visualization and Human Interpretability Evaluation component.

Retail Investors are the primary end users of the system. They use the dashboard to understand stock market trends, financial summaries, risk signals, and AI-generated explanations in a simplified and interpretable form.

Brokerage Firms can use the system as a value-added investor support tool to improve client engagement, investor education, and market understanding through explainable insight presentation.

Financial Analysts may use the dashboard to review summarized financial signals, contextual indicators, and investor-facing explanation methods that simplify technical outputs for wider audiences.

Fintech Platforms can adopt the proposed solution as an explainable financial insight module within digital investment or personal finance applications targeting beginner and intermediate investors.

Researchers and Developers are important stakeholders because they use the system for experimentation, comparative visualization analysis, evaluation, and future enhancement of financial insight presentation techniques.

System Administrators are responsible for maintaining the technical infrastructure, monitoring the availability of services, handling deployment-related issues, and ensuring secure and reliable operation of the platform.

5.2 Functional Requirements

The functional requirements describe the major services and operations that the system must provide in order to support investor-facing insight visualization and interpretability evaluation.

FR1: The system shall allow users to select a stock for analysis.

FR2: The system shall display market trend visualizations such as price movements, volume patterns, and related chart-based insight views.

FR3: The system shall display sentiment and event-based indicators derived from external market context and news-related analysis.

FR4: The system shall present simplified financial report summaries in an investor-friendly format.

FR5: The system shall display SHAP-based or feature-contribution explanation views for AI-generated predictions and insights.

FR6: The system shall provide risk indicators and warning mechanisms to highlight uncertainty, volatility, or caution-worthy market conditions.

FR7: The system shall support multiple explanation variants for selected insights, such as chart-only, chart-plus-text, chart-plus-feature-contribution, or chart-plus-event-context views.

FR8: The system shall allow users to interact with the dashboard and access integrated insight panels through a unified interface.

FR9: The system shall record user evaluation responses collected during interpretability and usability testing sessions.

FR10: The system shall compute and store interpretability-related metrics such as the Financial Insight Comprehension Score (FICS).

FR11: The system shall support the administration of usability and cognitive-load evaluation instruments such as SUS and adapted NASA-TLX.

FR12: The system shall store and present integrated insight summaries received from upstream analytical modules through a consistent dashboard structure.

FR13: The system shall support iterative refinement by enabling visualization variants to be compared and updated based on user evaluation findings.

5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes that the system must satisfy to ensure effective performance, usability, and reliability.

NFR1: Usability

The system shall be easy to learn and use for non-expert retail investors with limited financial and technical knowledge.

NFR2: Clarity

The system shall present information in a clear, understandable, and explanation-oriented manner using simple language, suitable visual structures, and minimal ambiguity.

NFR3: Accessibility

The system shall provide an interface that supports readability and ease of interaction for users with different levels of digital familiarity.

NFR4: Responsiveness

The dashboard shall load and update visualizations within an acceptable time to ensure smooth interaction and reduced user frustration.

NFR5: Security

The system shall protect stored data, user responses, and integrated insight records against unauthorized access and misuse.

NFR6: Reliability

The system shall consistently provide correct and available dashboard services during normal use and evaluation sessions.

NFR7: Maintainability

The platform shall be structured in a modular way so that visualization panels, explanation strategies, and evaluation features can be updated or improved easily.

NFR8: Scalability

The system shall support future expansion to additional stocks, insight types, visualization variants, and larger user groups without major redesign.

NFR9: Consistency

The interface shall maintain consistent layout, labeling, and interaction patterns across all dashboard panels and explanation variants.

NFR10: Interpretability Support

The system shall be designed to maximize user comprehension and minimize unnecessary cognitive load through investor-friendly visualization and explanation strategies.

These user requirements ensure that the proposed system is not limited to dashboard implementation alone, but functions as a human-centered investor support component that combines explainable visualization, multi-modal insight integration, and structured interpretability evaluation.

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system is **Colombo Stock Exchange retail investors**, especially **beginner and intermediate investors** who require understandable, trustworthy, and investor-friendly market insight support. Many of these users have access to digital trading platforms but lack the financial literacy needed to confidently interpret technical charts, AI-generated predictions, or multi-source financial signals. This makes them an ideal user group for a system that emphasizes explainability

and comprehension rather than raw analytics alone. This positioning is consistent with the component's purpose as the final investor-facing dashboard layer and with the proposal's focus on non-expert CSE investors.

A second important customer segment includes **brokerage firms and investment advisory services** that need better client-facing tools for presenting market information in an accessible and transparent form. In addition, **Sri Lankan fintech platforms and digital banking applications** represent a promising customer segment because they can embed explainable CSE insight visualizations as a value-added service within existing investment or wealth-support products. The proposal's current commercialization ideas already identify brokerage firms, fintech platforms, SEC/CSE-facing educational usage, and mobile investor-support contexts as practical adoption channels.

6.2 Value Proposition

The value proposition of the proposed system lies in its ability to convert **complex AI-generated financial outputs into explainable, investor-friendly, and localized visual insights**. Unlike conventional charting tools that mainly show raw prices, indicators, or technical signals, this system provides integrated visual explanations for stock trends, risk conditions, sentiment events, SHAP-based feature contributions, and financial report summaries within one dashboard. This directly aligns with the component's intended dashboard design, which includes five specialized panels supported by an interpretability evaluation framework.

A key differentiator is that the platform is not limited to displaying data; it is specifically designed to help non-expert investors understand **why** the system is showing a given insight. This makes the product useful not only as an information display tool, but also as a **decision-support and investor education platform**. The incorporation of human interpretability evaluation further strengthens the value proposition by ensuring that the dashboard is validated for real user comprehension rather than only technical correctness.

6.3 Revenue Model

A practical commercialization path for the system is a **freemium or subscription-based Software-as-a-Service (SaaS) model**. Under this model, a basic version of the dashboard can provide free access to standard trend charts, simple sentiment indicators, and general market summaries. Premium subscription tiers can unlock more advanced features such as explanation-rich visualizations, SHAP-based interpretability

panels, risk warning systems, enhanced financial report summaries, and deeper comparative insight views.

A second revenue model is **institutional licensing**. Brokerage firms, fintech providers, investment education services, and financial analytics providers can license the system for use within their own customer platforms or advisory services. A third possibility is an **API access model**, where selected visualization outputs or integrated insight summaries can be exposed to partner platforms through paid API usage. Similar commercialization directions are already reflected in the existing proposal material, which notes licensing to brokerage firms, client-facing use, and fintech integration as realistic opportunities.

6.4 Cost Estimation

The proposed system is relatively cost-effective because it is mainly built using **open-source technologies** such as React.js, Node.js, MongoDB Atlas, Recharts, D3.js, and Python. The current proposal also emphasizes that the system can be developed using a minimal-cost open-source stack.

The main cost areas for commercialization include:

- cloud hosting and deployment,
- database and storage services,
- maintenance of API integration pipelines,
- dashboard monitoring and security,
- user testing and support,
- and future scaling costs when supporting more users or more listed companies.

Since the initial research artefact is developed in an academic setting, early implementation costs remain relatively low. However, if the system is commercialized, operational expenses will increase with higher cloud usage, broader deployment, ongoing support, and future model integration. This is consistent with the cost-estimation pattern shown in comparable proposal structures, where development, cloud infrastructure, maintenance, and operational support are the main expenditure areas.

6.5 Pricing Strategy

The pricing strategy should remain **affordable for Sri Lankan retail investors** while also being sustainable for long-term product maintenance. A **tiered pricing model** is therefore the most suitable. A basic plan can provide free or low-cost access to simplified market insight visualizations. A premium plan can provide advanced interpretability features, detailed risk alerts, enhanced explanation variants, and richer financial summary analysis.

For institutional users such as brokerage firms, banks, fintech companies, or investor education providers, an **enterprise pricing model** can be used. Enterprise pricing may depend on user volume, dashboard usage level, integration complexity, or API access requirements. The friend's proposal uses a similar approach by suggesting free/basic access for general use and premium or enterprise packages for institutional adoption.

6.6 Competitive Advantage

The proposed system has several competitive advantages. First, it is specifically designed for the **Colombo Stock Exchange**, whereas many existing financial analytics platforms are optimized for large international markets and do not reflect the local context, investor behaviour, or interpretability needs of Sri Lankan users. The proposal and TAF both emphasize the need for a CSE-specific investor-facing solution.

Second, the system emphasizes **explainability and human interpretability**, not just charting or prediction output. It combines trend analysis, sentiment context, financial summaries, SHAP-based explanations, and risk indicators into a single investor-friendly workflow. Third, the system is strengthened by a **validated human-centered evaluation framework** using SUS, FICS, and NASA-TLX, which means the product is supported by empirical evidence of usability and comprehension rather than only software functionality. This gives it a meaningful edge over conventional dashboards that do not test whether users actually understand the presented insights.

6.7 Intellectual Property Considerations

Although the system uses open-source libraries and frameworks, the **overall system architecture, dashboard workflow, explanation strategy design, visualization logic, FICS-based evaluation method, and integration structure** can be treated as valuable intellectual property. Copyright protection may apply to the source code, interface

designs, dashboard layouts, evaluation instruments, and technical documentation developed as part of the project.

If the system is extended into a commercial product, additional protection mechanisms such as licensing agreements, restricted deployment models, proprietary integration strategies, and institutional use agreements may also be considered. At the same time, any external financial data, reports, or news sources used by the system must be handled in accordance with relevant licensing policies, data usage restrictions, and ethical standards. Comparable proposal material also treats the architecture, workflow, integrated logic, and implementation design as commercially protectable while acknowledging that underlying libraries remain open-source.

Overall, the commercialization potential of this system is strong because it addresses a practical and underserved need in the Sri Lankan financial market: delivering AI-generated stock market insights in a form that retail investors can actually understand, trust, and use.

BUDGET AND JUSTIFICATION

Table 2- Budget Estimation

Category	Item	Description	Estimated Cost (LKR)
Hardware	Development Computer Resources	Personal computers used for frontend, backend, preprocessing, testing, and documentation	20,000
Hardware	Storage Devices	Backup storage for outputs, recordings, and project files	10,000
Software	Development Tools/Utility Support	Plugins, utilities, debugging, documentation, and optional tool support	5,000
Cloud Services	Web Hosting and Compute	Hosting for dashboard and API services	30,000
Cloud Services	Data Storage	MongoDB Atlas, cloud storage for outputs and evaluation data	10,000
Human Effort	Development and Testing	Time spent by developers on system design, implementation, and testing	20,000
Miscellaneous	Documentation and Maintenance	Printing, maintenance, meeting coordination, and general project support	5,000

Total Estimated Budget = 100,000(LKR)

7.1 Hardware Costs

The proposed component does not require specialized high-end hardware such as GPU clusters or enterprise-grade servers because the main responsibility of this research is investor-facing dashboard development, insight fusion, visualization, and human interpretability evaluation. However, a reasonable level of computing resources is still required for frontend and backend development, prototype testing, API integration, local preprocessing, and result analysis. Personal computers will therefore be used for React.js dashboard development, Node.js backend integration, Python-based preprocessing, metric computation, and documentation activities. This is consistent with the current proposal, which identifies development computers and storage as essential cost items.

In addition, removable or external storage may be required to maintain project backups, evaluation materials, recordings, generated outputs, and intermediate datasets. Since the system handles multiple AI outputs and user evaluation artifacts, backup storage is a reasonable supporting expense. The current proposal also includes storage and backup-related costs within the budget framework.

7.2 Software Licenses

Most of the software and development libraries used in this component are open source, which significantly reduces the financial burden of implementation. Technologies such as React.js, Node.js, Express.js, MongoDB Atlas free tier, Python, Recharts, D3.js, and GitHub are either free or available under academic-friendly usage models. The current proposal and feasibility analysis both emphasize that the system relies primarily on open-source tools and community-supported frameworks.

However, a small software-related allocation is still reasonable for supplementary tools, extensions, plugins, or utilities that may support development, debugging, security monitoring, documentation, or deployment preparation. In addition, optional premium prototyping or collaboration features may be used if required during later refinement stages. Therefore, a modest software license or development tool cost is justified even though the primary stack itself remains open source.

7.3 Cloud Services

Cloud services are a key cost category because the proposed component depends on web deployment, API communication, cloud-based persistence, and multi-module integration. The dashboard frontend may be deployed using free-tier or low-cost hosting options such as Vercel or Firebase, while the Node.js integration layer and MongoDB Atlas database support storage and retrieval of dashboard-ready insight data. The current proposal already notes the use of Firebase or Vercel for hosting, MongoDB Atlas for database services, and free-tier cloud support for prototype deployment.

Cloud resources may also be required for storing AI outputs, session records, evaluation responses, and documentation backups. In addition, security and monitoring-related costs such as SSL support, error logging, and API performance monitoring may arise during implementation or deployment. The existing budget in the current proposal already includes cloud database, cloud storage, security, and monitoring as practical operational expenses.

7.4 Human Effort Cost

Although this work is conducted as an academic research project, human effort is still a major cost factor in terms of implementation time, evaluation coordination, refinement, and documentation. This component requires not only technical dashboard development but also repeated interpretability testing through formative and summative evaluation. The methodology and work plan show that the component includes wireframing, API contract definition, prototype development, React dashboard development, fusion service integration, formative evaluation, summative evaluation, refinement, and documentation across multiple project phases.

An additional human-centered cost arises from participant-based evaluation. Since the project includes formative think-aloud sessions with around 5–8 users and summative usability sessions with around 15–20 users, participant incentives, coordination time, and evaluation logistics become part of the effective research cost. The current proposal explicitly includes participant incentive costs as one of the major human-centered evaluation expenses.

7.5 Justification for Expenses

The budget is justified because it directly supports the main technical and research activities of the component. Hardware costs are necessary for development, preprocessing, integration, and secure storage of outputs. Software-related costs remain low because the core technology stack is open source, but minor supporting tools may still be needed. Cloud service costs are essential because the system relies on dashboard hosting, API communication, database access, and storage of integrated multi-modal outputs. Human effort cost is justified not only by implementation work, but also by the structured human interpretability evaluation process, which includes participant management, usability sessions, metric computation, and post-evaluation refinement. The current feasibility analysis also supports this budget direction by confirming that the project is economically feasible through reliance on free-tier cloud services and open-source technologies, while identifying internet connectivity, LLM/API-related usage, and usability testing as the main non-zero costs.

Overall, the proposed budget is cost-effective because it covers the essential resources required to design, implement, test, refine, and document the Insight Visualization and Human Interpretability Evaluation component without exceeding the typical financial scope of an undergraduate research project.

WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) organizes the activities of the **Insight Visualization and Human Interpretability Evaluation** component into clearly defined phases and tasks. The purpose of the WBS is to ensure that the work is manageable, logically sequenced, and aligned with the overall group project timeline. Since this component depends on outputs from upstream AI modules and also contributes the final investor-facing layer of the system, the workload must be distributed in a way that balances **technical implementation, integration work, human-centered evaluation, and documentation activities**. This is consistent with both the TAF expectations and the current component methodology.

The work is divided across six major phases. Early phases focus on requirement understanding, teammate coordination, API contract planning, and wireframe preparation. Middle phases focus on prototype development, React dashboard implementation, and multi-modal integration. Later phases focus on interpretability evaluation, interface refinement, validation, and documentation. This phased structure ensures that development begins even before all upstream modules are fully completed by using agreed mock data and formalized JSON contracts.

Because this is an individual component within a larger group project, the WBS also acknowledges interactions with the other team members. Outputs from the **ML and XAI module, financial document understanding module, and sentiment/event-driven module** are integrated into this component through the Insight Fusion Service, while this component is responsible for turning those outputs into an interpretable dashboard and evaluating whether end users can understand them.

8.1 Justification for Expenses

Although this report focuses on the **Insight Visualization and Human Interpretability Evaluation** component, the WBS must reflect coordination with the other project members because the dashboard depends on upstream outputs.

- **S.A.H. Samarathunga (IT21296482)** – Provides ML trend analysis, XAI outputs, and model-driven market insights for downstream dashboard integration.
- **K.C.S. Fernando (IT21471162)** – Provides financial report understanding outputs and LLM-generated financial summaries for dashboard cards and explanation panels.
- **Harithra K. (IT22065308)** – Provides sentiment scores, event classifications, and contextual market signals for sentiment and event-driven visualization panels.
- **T.G.R.N.S. Wijesooriya (IT22547088)** – Responsible for insight fusion, dashboard visualization, explanation-oriented presentation, human interpretability evaluation, and iterative refinement of the final investor-facing interface.

Table 3- Work Breakdown Structure Table

WBS ID	Phase	Task Description	Responsible Member	Expected Output
1.0	Initiation	Review TAF, supervisor feedback, and project scope	IT22547088	Component scope finalized
1.1	Initiation	Conduct literature review on financial visualization, XAI, and usability evaluation	IT22547088	Literature base and research direction
1.2	Initiation	Gather integration requirements from teammate modules	IT22547088 + team	Input specifications collected
2.0	Planning and Design	Define dashboard requirements and insight categories	IT22547088	Requirement specification
2.1	Planning and Design	Define API contracts and expected JSON schemas for upstream outputs	IT22547088 + team	API contract draft
2.2	Planning and Design	Design low-fidelity wireframes in Figma	IT22547088	Initial UI mockups
2.3	Planning and Design	Prepare evaluation instruments (SUS, FICS, NASA-TLX, task scripts)	IT22547088	Evaluation materials
3.0	Implementation	Create mock data and rapid prototype for early validation	IT22547088	Prototype for formative testing
3.1	Implementation	Develop Node.js / Express insight fusion service	IT22547088	Fusion backend service
3.2	Implementation	Develop React.js dashboard panels	IT22547088	Investor-facing dashboard UI
3.3	Implementation	Integrate ML, LLM, and sentiment outputs into dashboard	IT22547088 + team	Unified multi-modal dashboard
4.0	Testing and Evaluation	Conduct formative think-aloud evaluation sessions	IT22547088	Early usability findings

WBS ID	Phase	Task Description	Responsible Member	Expected Output
4.1	Testing and Evaluation	Conduct summative evaluation using SUS, FICS, NASA-TLX	IT22547088	Quantitative evaluation results
4.2	Testing and Evaluation	Analyze task failures, cognitive load, and comprehension barriers	IT22547088	Evaluation analysis report
5.0	Refinement	Improve visualization panels and explanation strategies	IT22547088	Refined dashboard
5.1	Refinement	Validate post-refinement improvements	IT22547088	Updated metrics and findings
6.0	Documentation	Prepare technical documentation and API references	IT22547088	Technical documentation
6.1	Documentation	Prepare proposal, status reports, final report, and presentation material	IT22547088	Dissertation deliverables

8.2 Timeline Alignment

The WBS is aligned with the full project lifecycle from the early planning stage to final validation and reporting. The current proposal already organizes this component across six phases from **January to December**, with initiation and planning tasks occurring first, followed by implementation, evaluation, refinement, and documentation.

A practical timeline alignment for this component is as follows:

- **Phase 1 – Initiation:** Literature review, TAF review, and requirement gathering
- **Phase 2 – Planning and Design:** Wireframes, API contracts, evaluation preparation
- **Phase 3 – Implementation:** Prototype development, dashboard coding, fusion service integration
- **Phase 4 – Testing and Evaluation:** Formative and summative interpretability testing
- **Phase 5 – Refinement:** Dashboard improvement and post-refinement validation

- **Phase 6 – Documentation:** Final report writing, technical documentation, and presentation preparation

This sequence is well aligned with the component methodology, which begins with API contract definition and mock data preparation, moves into wireframing and rapid prototyping, continues into React dashboard development, and concludes with interpretability evaluation and refinement.

8.3 Balanced Workload Distribution

The workload distribution is balanced by separating the component into **planning**, **technical development**, **evaluation**, and **documentation** stages. This prevents the work from being concentrated only on coding or only on report writing. Instead, the component progresses through a structured research-development cycle where each phase contributes to the next.

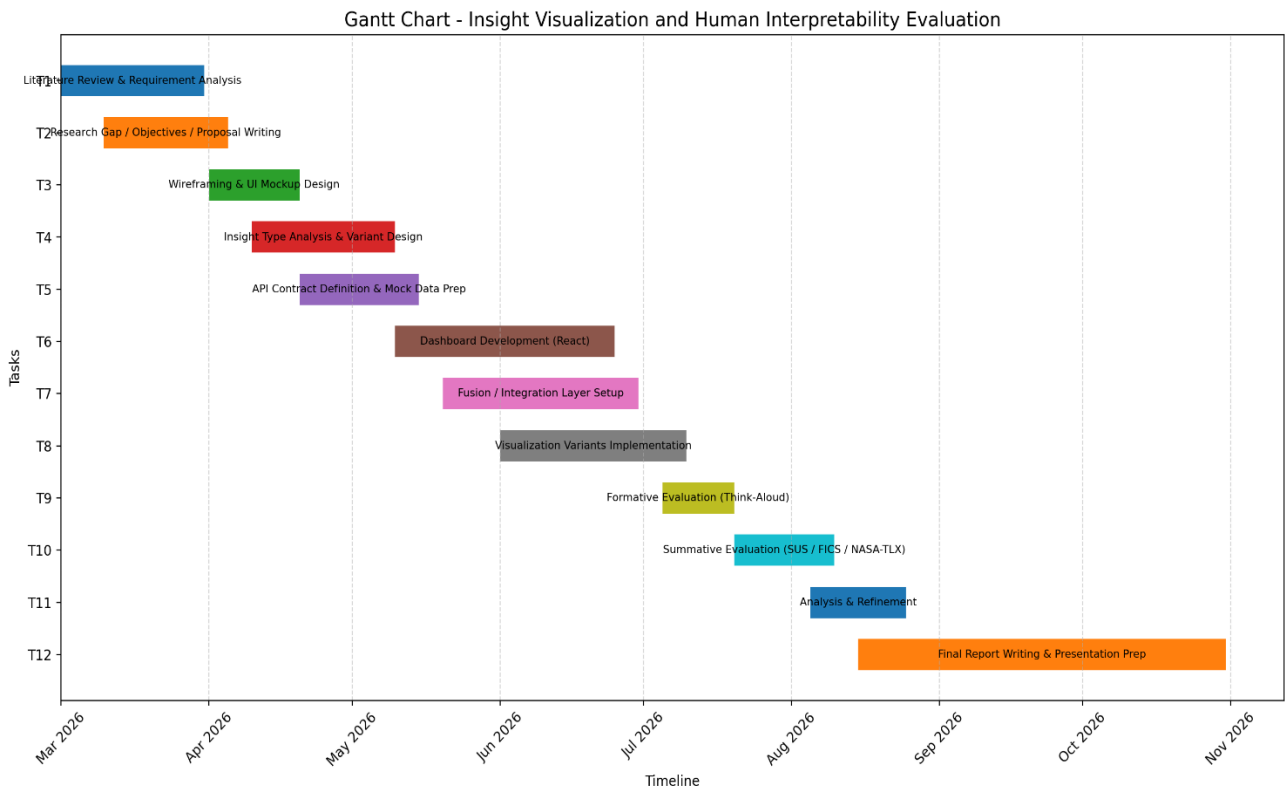
The design also reduces dependency risk by allowing dashboard and prototype development to begin with **mock data** before all upstream modules are fully completed. At the same time, final integration is scheduled only after API contracts and expected data structures are agreed upon with teammates. This helps maintain balance between independent progress and team-level coordination.

In addition, workload balance is maintained by ensuring that:

- early phases focus on research and design,
- middle phases focus on implementation and integration,
- later phases focus on evaluation and refinement,
- and the final phase focuses on reporting and presentation.

This distribution is especially appropriate for the **Insight Visualization and Human Interpretability Evaluation** component because it includes both engineering work and human-centered research work. The TAF explicitly expects this component to handle dashboards, multi-modal UI integration, interpretability evaluation, and usability-driven refinement, making a balanced WBS essential.

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